

Optimization of the MobileNetV4 for Heritage Site Classification Using Post-Training Quantization

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Abstract. Vision Transformers and Deep Neural Networks have been widely used for image classification task but deploying them on resource-constrained edge devices presents significant challenges due to their high computational demands. To address this, we employed the lightweight MobileNetV4 model, enhanced with advanced features such as the Universal Inverted Bottleneck and Mobile Multi-Query Attention, for heritage site classification on the Indian Heritage Digital Space Dataset. A comparative analysis was conducted between MobileNetV4 and MobileNetV3, both trained on the IHDS dataset, by applying 4-bit and 6-bit Post-Training Quantization. MobileNetV4, with an original model size of 33.99 MB, was compressed to 5.38 MB using 6-bit quantization and further reduced to 4.05 MB with 4-bit quantization, achieving accuracies of 91.47% and 88.62%, respectively. In contrast, MobileNetV3, with an original model size of 20.9 MB, experienced greater accuracy loss after quantization, with 6-bit and 4-bit accuracies of 87.50% and 82.30%, and corresponding model sizes of 4.39 MB and 2.68 MB. These results demonstrate that MobileNetV4 outperforms MobileNetV3 in maintaining accuracy post-quantization while achieving significant model size reduction. This highlights MobileNetV4 as an efficient and practical choice for deploying heritage site classification models on edge devices.

Keywords: Vision Transformers, Deep Neural Networks, MobileNetV4, Post-Training Quantization, IHDS Dataset, Indian Cultural Heritage, Efficient Deployment.

1 Introduction

India's rich cultural heritage is showcased in its iconic heritage sites such as the Taj Mahal, Khajuraho temples, Red Fort, Hampi, Pattadakal monuments, etc. These sites are not just architectural wonders but also embody India's historical narratives, cultural evolution, and religious diversity. Preserving these cultural heritage sites is essential for preserving the history embedded in their architecture. Tasks such as image classification will help in accurately categorizing

and analyzing images, which will assist in documenting architectural details, detecting damage, and guiding restoration with precision.

As we enter the age of Artificial Intelligence (AI) [1], new technologies such as Vision Transformers (ViTs) [2, 3], and Deep Neural Networks (DNNs) [4, 5] offer a novel approach to image classification.

ViTs work by dividing images into patches and using self-attention to capture complex patterns. However, they require extensive data and computational power, which makes them difficult to deploy on edge devices. Similarly, DNNs focus on local details, missing broader patterns, and also require significant processing power, which limits their use on smaller devices.

In Table 1, we present a comparative analysis of popular deep learning models including ViTs, ResNet-50, and VGG-19, based on their parameter count and model size. While these architectures have exhibited high performance in various image recognition tasks, their computational and memory demands hinder their effectiveness in resource-constrained environments like edge devices. To address this limitation, we extended our analysis to include MobileNet architectures [6, 7], which are lightweight and optimized for mobile and edge computing.

Table 1: Comparison of Model Size and Parameters for Various Architectures

Metric	ResNet 50	VGG 19	ViT Base
Model Size (MB)	98	549	344
Parameters (M)	25.6	143.7	86

Over the years, multiple MobileNet architectures, such as MobileNetV2 [8, 9] and MobileNetV3 [10, 11], have been developed to support efficient deployment and image classification on edge devices. MobileNetV2 introduced new concepts like inverted residuals and linear bottlenecks, reducing computational cost and optimizing the model for efficient deployment onto edge devices. However, MobileNetV2 occasionally struggled with resource efficiency on lower-power devices. Building on this, MobileNetV3 introduced refinements such as automated Neural Architecture Search (NAS) [12, 13, 14] and Squeeze-and-Excitation (SE) modules [15, 16], which boosted both accuracy and efficiency beyond MobileNetV2. Despite these advancements, the reliance on manually crafted design elements limited flexibility for rapidly evolving AI applications.

The latest advancement in these architectures is MobileNetV4 [17] which introduces the Universal Inverted Bottleneck(UIB), an enhanced version of MobileNetV2’s Inverted Bottleneck(IB), featuring additional depthwise convolutions and advanced mixing techniques. It also integrates Mobile Multi-Query Attention(MQA) to accelerate attention mechanisms and employs an improved NAS to optimize layer configurations for diverse hardware. These advancements enhance flexibility, reduce latency, and boost performance, making MobileNetV4 faster and more versatile for edge device deployment.

Leveraging these innovations, MobileNetV4 achieves a drastic reduction in model size. With Post-Training Quantization (PTQ), the model size decreases from 33.99 MB to 6.37 MB with 6-bit PTQ, and further to 4.25 MB with 4-bit PTQ, while maintaining high accuracy levels of 91.47% and 88.62%, respectively. This combination of compact size and robust performance makes MobileNetV4 highly efficient for resource-constrained environments.

The proposed workflow as shown in Fig 1 begins by introducing the task of image classification and addressing the challenges of deploying ViTs and DNNs on edge devices due to high computational demands in Section 1. Section 2 reviews related work on MobileNet architectures and the role of PTQ in achieving efficient deployment without significant accuracy loss. Section 3 elaborates on the MobileNetV4 architecture, highlighting its advantages over its predecessors. Section 4 presents the proposed approach, where the MNv4-Conv-M variant is trained on the IHDS dataset and optimized using 4-bit and 6-bit PTQ. Section 5 analyzes the results and provides a comparative analysis between MobileNetV4 and MobileNetV3 architectures, based on reduction in model size and accuracy trade-offs, while Section 6 concludes by demonstrating the effectiveness of MobileNetV4 with PTQ for efficient edge deployment.

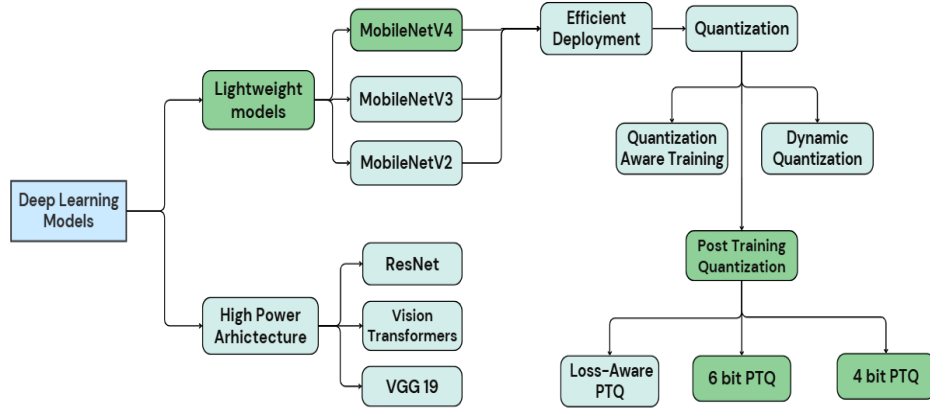


Fig. 1: Proposed WorkFlow.

2 Background Study

Recent advancements in deep learning have enabled efficient image classification, particularly for resource-constrained environments like edge devices. Various

studies have explored techniques such as transfer learning and PTQ to optimize model performance and memory usage. This section reviews notable works related to cultural heritage site classification and innovative quantization approaches for neural networks highlighting their methodologies, achievements, and relevance to the current study.

Kulkarni et al. [18] explored the use of advanced deep-learning techniques for image classification of Indian cultural heritage sites. It proposes a crowd-sourcing framework under the IHDS project to facilitate the preservation and documentation of heritage information. The authors created a dataset of 108,080 images spanning 78 heritage sites, sourced through manual collection and crowd-sourcing. They utilized a transfer learning approach using a pre-trained MobileNetV2 model, achieving a classification accuracy of 98.75%, outperforming traditional methods like Simple Multiclass and Ensemble of Binary Classifiers. The framework includes a web-based platform for image uploading, classification, and retrieval, fostering engagement from historians, tourists, and researchers. The authors highlight the scalability of their approach and propose further optimization through model quantization and potential applications in 3D reconstruction.

Banner et al. [19] introduced a 4-bit PTQ method for Convolution Neural Networks(CNN), which enables model compression without requiring the full training dataset or additional retraining. The authors propose three techniques to minimize quantization errors, viz., Analytical Clipping for Integer Quantization(ACIQ) to limit activation ranges, a per-channel bit allocation strategy to optimize bit-width allocation across channels, and bias correction to adjust quantization bias in weights. Together, these methods allow for accurate 4-bit quantization, achieving close to full-precision accuracy on standard models like ResNet and VGG. The approach is intended to facilitate the rapid deployment of neural networks with reduced memory and computation requirements.

Nahshan et al. [20] introduce Loss Aware Post-Training Quantization(LAPQ), a novel approach designed to enhance neural network quantization for deployment on resource-constrained devices. It identifies and addresses limitations with current PTQ approaches, particularly for low-bit configurations, due to non-separable loss landscapes and steep curvatures. The proposed methodology combines layer-wise optimization, quadratic approximation, and joint gradient-free optimization to reduce quantization errors while maintaining high accuracy effectively. Experimental results show that LAPQ’s superior performance over existing methods, achieving nearly full-precision baseline accuracy even for 4-bit quantization. Its straightforward implementation and compatibility with standard hardware make it a valuable solution for optimizing models on edge devices.

3 MobileNetV4 Architecture

With an emphasis on mobile efficiency, MobileNetV2 and MobileNetV3 reflect significant advancements in the design of mobile-friendly deep learning models, each aimed to improve speed and efficiency without compromising accuracy.

MobileNetV2 introduced IB layers, which reduced computational costs and improved latency. However, its design was not compatible across various hardware types and also struggled with complex attention mechanisms, which in turn limited its accuracy for certain tasks. MobileNetV3 addressed some of these issues by introducing hardware-aware NAS and SE layers to further boost speed on mobile CPUs. Despite these improvements, MobileNetV3 models were primarily optimized for specific hardware, resulting in inconsistent performance across a range of mobile processors.

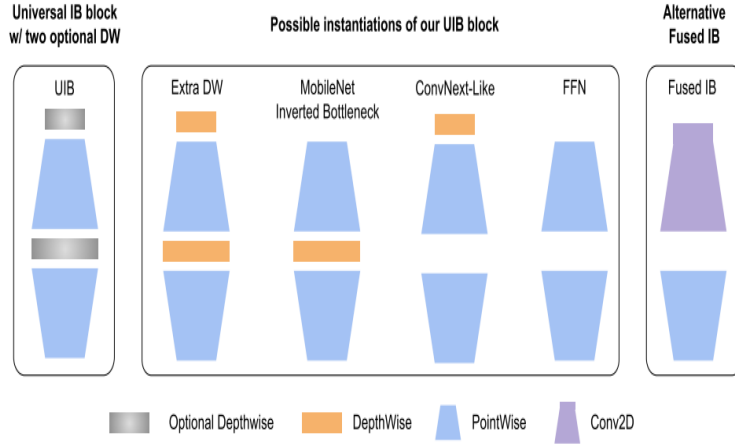


Fig. 2: Architecture of Universal UIB Block for MobileNetV4 [17].

After all this progress, MobileNetV4 was developed to overcome the remaining limitations of MobileNetV2 and V3 by introducing innovations that enhance both flexibility and efficiency across a wide range of mobile hardware. The MobileNetV4 architecture consists of 4 UIB combinations as shown in Fig 2: ExtraDW, ConvNext, IB, and FFN. These combinations highlight the importance of using different instantiation blocks at various stages of the network. The design of ExtraDW helps to enlarge the receptive field and enhance spatial mixing, thus effectively reducing resolution loss. Similarly, ExtraDW is frequently selected in the early stages of MNv4-Conv models for the same reasons. For the final layers, where preceding layers have conducted substantial spatial mixing, FFN and ConvNext are chosen because channel mixing provides a larger incremental gain.

Additionally, the Mobile MQA block is introduced which enhances the attention score calculation as shown in Eq. (1), which improves attention mechanisms for mobile devices by reducing memory requirements. MQA accomplishes this by sharing information across multiple attention heads, achieving over a 39% speedup on accelerators without sacrificing accuracy.

$$\begin{aligned} \text{Mobile_MQA}(\mathbf{X}) &= \text{Concat}(\text{attention}_1, \dots, \text{attention}_n) \mathbf{W}^O \\ \text{attention}_j &= \text{softmax} \left(\frac{(\mathbf{X} \mathbf{W}^{Q_j})(\text{SR}(\mathbf{X}) \mathbf{W}^K)^T}{\sqrt{d_k}} \right) (\text{SR}(\mathbf{X}) \mathbf{W}^V) \end{aligned} \quad (1)$$

where: W_{Qj} : Query weight matrix for the j -th attention head; W_K : Key weight matrix; W_V : Value weight matrix; SR: Spatial Reduction; d_k : Dimensionality of the query/key vectors.

To ensure universal efficiency, MobileNetV4 leverages the Roofline Model [21], which helps identify hardware bottlenecks by modeling the relationship between memory bandwidth and computational demands across devices. The architecture also uses a refined two-stage NAS process, which carefully balances resource usage across various devices through a combination of coarse-grained and fine-grained searches, resulting in an adaptable model that is both highly accurate and resource-efficient. These innovations establish MobileNetV4 as a robust and versatile solution within the mobile ecosystem, offering performance scalability and consistent accuracy across a wide array of mobile and edge hardware.

4 Proposed Work

The primary focus of this work is to develop a robust deep learning model for image classification on the IHDS dataset, ensuring efficient deployment on edge devices without compromising accuracy. The proposed approach integrates the MNv4-Conv-M architecture with the PTQ technique to balance accuracy and model size effectively.

The IHDS dataset is divided into training, validation, and testing subsets, with data augmentation techniques like rotations, flips, and normalization used to address class imbalances and improve generalization. The MNv4-Conv-M, a MobileNetV4 variant, is initialized with pre-trained ImageNet weights to accelerate training, and a custom classification layer is fine-tuned for optimal performance on the IHDS dataset. Dropout regularization [22, 23] is applied to enhance robustness and mitigate overfitting.

The model is trained using the Adam optimizer [24] and cross-entropy loss [25], as defined in Eq. (2), with dynamic learning rate adjustments ensuring stability and efficiency. This approach achieves high accuracy and compact model size, making it ideal for resource-constrained edge devices.

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c}), \quad (2)$$

where: N : Number of samples; C : Number of classes; $y_{i,c}$: Ground truth (1 if sample i belongs to class c , else 0); $p_{i,c}$: Predicted probability for sample i being in class c .

To ensure the smooth deployment of the model, we applied PTQ. This PTQ is implemented as a custom-written method that quantizes the Conv2D layers

of the model. This technique compresses the model by reducing the precision of its weights, as shown in Eqs. (3) and (4), while maintaining acceptable accuracy levels.

$$w_q = \text{round} \left(\frac{w - \min(w)}{\Delta} \right) \cdot \Delta + \min(w), \quad (3)$$

where w_q is the quantized weight and w is the original weight. The quantization step size Δ is given by:

$$\Delta = \frac{\max(w) - \min(w)}{2^k - 1}, \quad (4)$$

where k is the bit precision (e.g., 4-bit or 6-bit), and $\min(w)$ and $\max(w)$ are the weight range limits.

In our work, as shown in Fig 3, we applied PTQ to the Conv2D layers of the MobileNetV4 architecture, specifically the MNv4-Conv-M variant, to classify heritage sites.

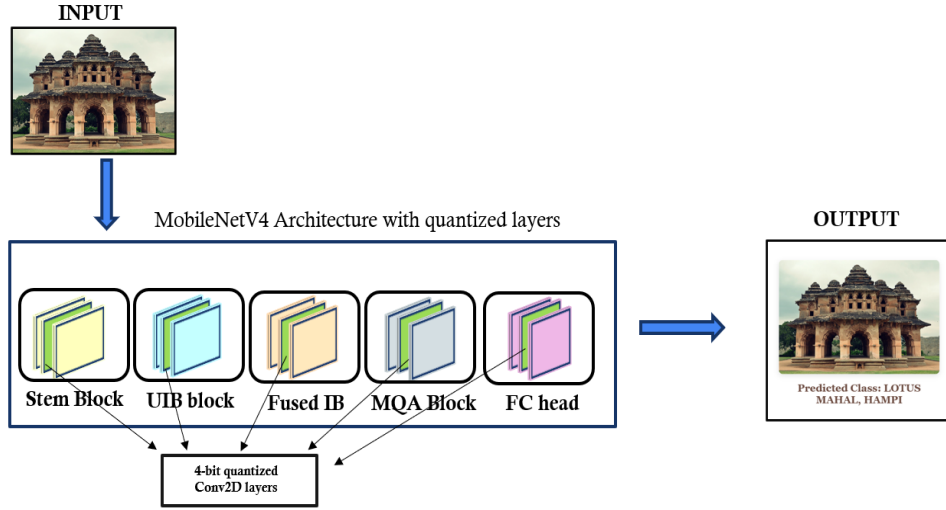


Fig. 3: Proposed Architecture.

We focused on quantizing the Conv2D layers of the model to 4-bit precision and then to 6-bit precision, performing a comparative analysis of the quantized models in terms of size and accuracy. Classification accuracy is tested to check the retention of original performance post-quantization. To evaluate the similarity between feature maps before and after quantization, the Cosine Similarity is calculated as shown in Eq. (5):

$$\text{Cosine Similarity} = \frac{\sum_{i=1}^n f_{1,i} f_{2,i}}{\sqrt{\sum_{i=1}^n f_{1,i}^2} \sqrt{\sum_{i=1}^n f_{2,i}^2}}, \quad (5)$$

where f_1 and f_2 are the feature maps (original and quantized). Overall this work presents an efficient way to deploy the MobileNetV4 model onto edge devices with acceptable classification accuracy and significantly reduced model size.

5 Results and Analysis

This section provides a brief overview of the IHDS dataset and presents a comparative analysis of the MobileNetV4 and MobileNetV3 architectures to evaluate the impact of PTQ on model size and accuracy

5.1 Dataset Description

The IHDS dataset, as shown in Fig. 4, contains images of notable Indian monuments such as the Agoreshwar Temple, Aihole Temple, Chandramouleshwar Temple, and the Hampi Chariot, which highlight India's rich architectural and cultural heritage. This dataset comprises 133,481 images, categorized into 50 distinct classes. The research work leveraging this dataset is supported under the Indian Heritage in Digital Space (DST/ICPS/IHDS/2018) initiative, a part of the Interdisciplinary Cyber Physical Systems Programme funded by the Department of Science and Technology, Government of India.



Fig. 4: Sample Images of Heritage Sites from the IHDS dataset

5.2 Experimental Results and Analysis

In our work, we used the IHDS dataset to evaluate the performance of PTQ on MobileNetV4 and MobileNetV3 models. The results demonstrate a trade-off between model size and accuracy after applying 6-bit and 4-bit PTQ separately.

As shown in Table 2, a comparison of the MobileNetV3 and MobileNetV4 architectures highlights their reduced model sizes after PTQ. MobileNetV4, with an original model size of 33.99 MB, was reduced to 5.38 MB with 6-bit quantization and further to 4.05 MB with 4-bit PTQ. On the other hand, MobileNetV3, with an original size of 20.90 MB, was reduced to 4.39 MB with 6-bit PTQ and to 2.68 MB with 4-bit PTQ. Although MobileNetV3 achieved a greater reduction in model size, there was a significant drop in its accuracy compared to MobileNetV4, as shown in Table 3. The accuracy comparison as shown in Table 3 highlights the effectiveness of PTQ in reducing model size while maintaining acceptable accuracy. MobileNetV4’s original accuracy was 95.84%, compared to MobileNetV3’s 93.00%. After 6-bit PTQ, MobileNetV4’s accuracy decreased slightly to 91.47%, while MobileNetV3 experienced a more significant drop to 87.50%. With 4-bit PTQ, MobileNetV4 maintained a competitive accuracy of 88.62%, while MobileNetV3’s accuracy dropped further to 82.30%. These results indicate that MobileNetV4 retains higher accuracy at lower precision levels as compared to MobileNetV3. Overall, the MobileNetV4 model trained on

Table 2: Comparison of Model Sizes in Megabytes (MB)

Model	Original Size	6-bit Quantized	4-bit Quantized
MobileNetV4	33.99	5.38	4.05
MobileNetV3	20.90	4.39	2.68

Table 3: Comparison of Accuracy (%)

Model	Original Accuracy	6-bit Quantized	4-bit Quantized
MobileNetV4	95.84	91.47	88.62
MobileNetV3	93.00	87.50	82.30

the IHDS dataset delivers robust and reliable performance, even after quantization, ensuring efficient deployment on edge devices. The substantial reduction in model size after quantization makes deployment not only easier but also significantly more efficient. These results demonstrate that an ideal balance between model size and accuracy can be achieved, thereby fulfilling the objective of this study to ensure efficient deployment on edge devices with strong performance.

5.3 Screenshots

The proposed work is demonstrated through a web interface, as shown in Fig. 5, to facilitate user interaction. This interface is designed using the Flask framework. Users can upload an image, and the module classifies the uploaded image into the appropriate class labels.

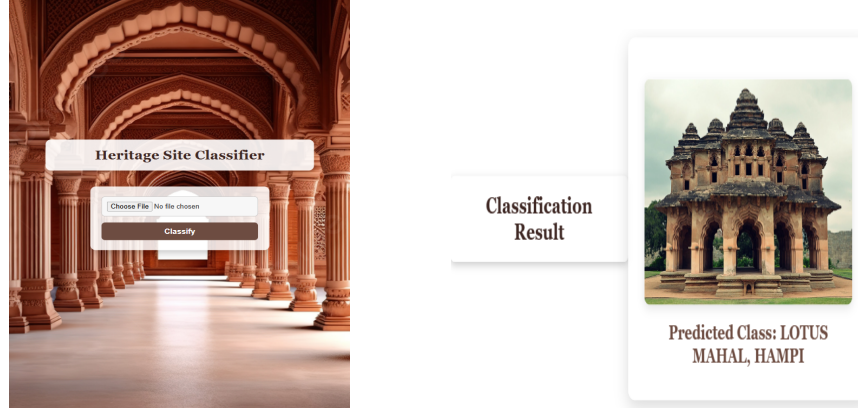


Fig. 5: Sample web portal for uploading an image and predicting the heritage site.

6 Conclusion and Future Scope

This research demonstrates the effectiveness of MobileNetV4 for efficient deployment on edge devices using PTQ. Beyond cultural heritage, MobileNetV4's lightweight and efficient architecture can be applied to various fields such as medical imaging, autonomous vehicles, and real-time object detection in mobile applications. With minimal accuracy loss and significant size reduction, MobileNetV4 offers a scalable solution for image classification in resource-constrained environments. Future work could explore further optimizations and broader applicability across diverse datasets and hardware platforms.

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